

Higher derivatives



1 Suppose $y = \frac{2}{x^5}$. Find:

a $\frac{dy}{dx}$

b $\frac{d^2y}{dx^2}$

c $\frac{d^3y}{dx^3}$

2 Suppose $y = 5x^8$. Find:

a $\frac{dy}{dx}$

b $\frac{d^2y}{dx^2}$

c $\frac{d^3y}{dx^3}$

d $\frac{d^4y}{dx^4}$

e $\frac{d^5y}{dx^5}$

3 Consider the function $y = -2x^4 + 4x^3 - 7x^2 + 8x - 9$. Find:

a $\frac{dy}{dx}$

b The value of $\frac{dy}{dx}$ when $x = 3$

c $\frac{d^2y}{dx^2}$

d The value of $\frac{d^2y}{dx^2}$ when $x = 4$

e $\frac{d^3y}{dx^3}$

f The value of $\frac{d^3y}{dx^3}$ when $x = 6$

4 Find the second derivative of the following functions:

a $y = x^4 - 6x^3 + 8x^2 - 10$

b $f(x) = x^{\frac{1}{7}}$

c $f(x) = (3x + 2)^4$

d $f(x) = \frac{4}{(x + 8)^2}$

e $y = (7 - 6x^2)^{\frac{1}{3}}$

f $y = 4\sqrt{6 - x}$

g $f(x) = x^2(x^3 + 3)$

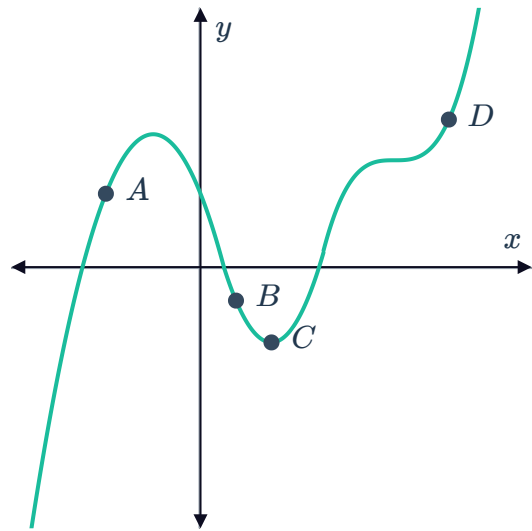
h $f(x) = (x^4 + 2)(3 - 2x^3)$



Concavity

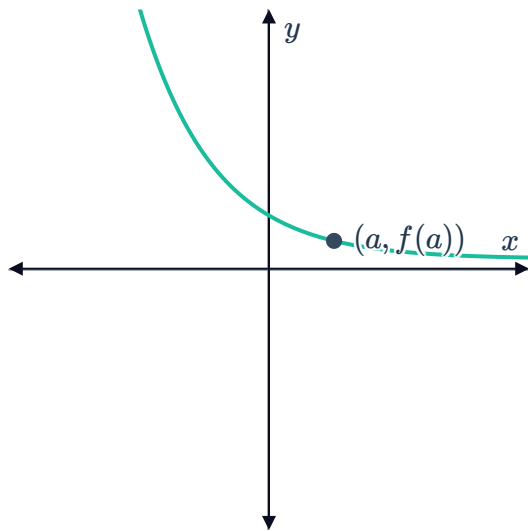
5 Consider the given function $y = f(x)$ with points A, B, C and D :

- a At which of these points is $f''(x) > 0$?
- b At which of these points is $f''(x) < 0$?

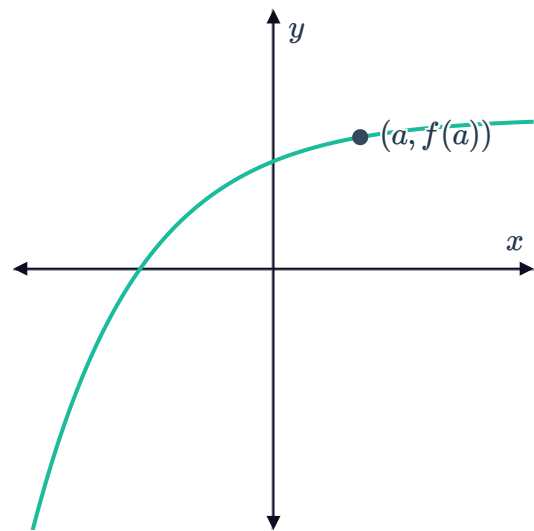


6 For each of the following curves, state the sign of $f'(a)$ and $f''(a)$:

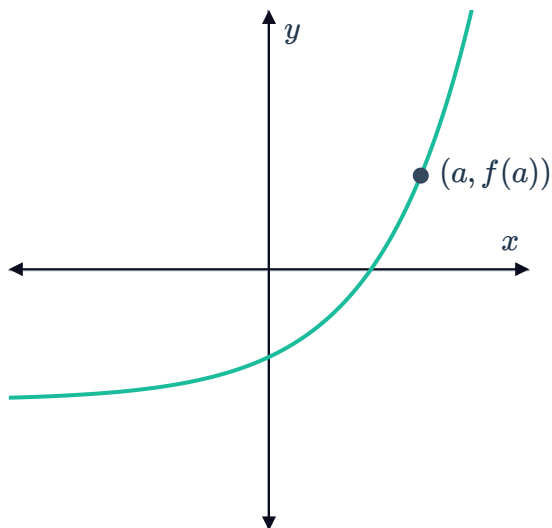
a



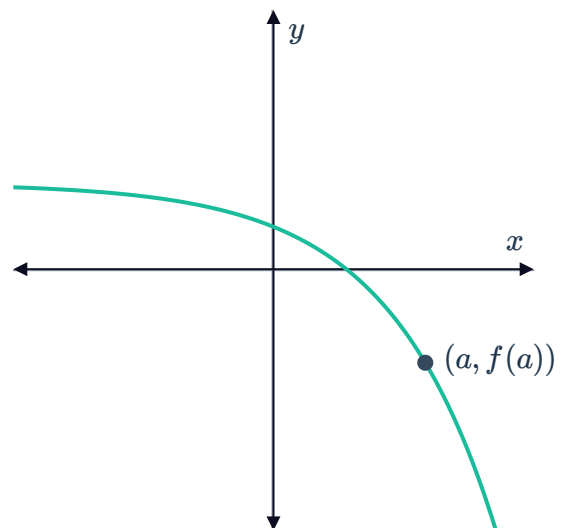
b



c



d





7 For each of the following quadratic functions:

- i** State whether the leading coefficient of $f(x)$ is positive or negative.
- ii** Hence, determine the nature of the turning point.
- iii** Find $f'(x)$.
- iv** Find $f''(x)$.
- v** State whether the curve is concave up or down.

a $f(x) = x^2 - 4x + 9$

b $f(x) = -x^2 + 4x - 9$

8 Consider the function $f(x) = 4x^2 + 3x + 2$.

- a** Find the first derivative.
- b** Find the second derivative.
- c** Complete the following tables of values:

x	0	1	2	3	4
$f'(x)$	3				

- d** Describe the rate of change of the gradient.
- e** Are there any points of inflection on $f(x)$? Explain your answer.

9 Consider the function $f(x) = -3x^2 + 12x + 2$.

- a** Find $f'(x)$.
- b** Find $f''(x)$.
- c** Describe the rate of change of the gradient.
- d** Are there any points of inflection on $f(x)$? Explain your answer.

10 For each of the following functions:

- i** State the values of x for which the graph of the function is concave up.
- ii** State the values of x for which the graph of the function is concave down.
- iii** Explain why there is no point of inflection.

a $y = \sqrt{x+1}$

b $y = -\sqrt{x-2}$

c $y = \frac{1}{x}$

d $y = \frac{1}{x^2}$

11 Consider the function $y = (x + 6)^3$.



- a State the transformation that turns $y = x^3$ into $y = (x + 6)^3$.
- b Find the point of inflection of $y = x^3$.
- c Find the point of inflection of $y = (x + 6)^3$.

d Complete the following table of values:

x	-7	-6	-5
y'			
y''		0	

- e Is the point of inflection, a horizontal point of inflection? Explain your answer.
- f For what values of x is the graph concave up?

12 Consider the function $y = (x + 5)^3$.

- a State the transformation that turns $y = x^3$ into $y = (x + 5)^3$.
- b Find the point of inflection of $y = x^3$.
- c Find the point of inflection of $y = (x + 5)^3$.

d Complete the following table of values:

x	-6	-5	-4
y'			
y''		0	

- e Is the point of inflection, a horizontal point of inflection? Explain your answer.
- f For what values of x is the graph concave up?

13 Consider the function $y = x^4 - 8x^3 - 9$.

- a Find y'' .
- b Find the points of inflection.

c Complete the following table of values:

x	-2	0	2	4	6
y'					
y''		0		0	

- d Classify each point of inflection as an ordinary or horizontal point of inflection.
- e For what values of x is the graph concave up?

14 Consider the function $y = 4x^3 - 16x^2 + 4x + 6$.

- a Find y'' .
- b Find the point of inflection.
- c Is the point of inflection, a horizontal or an ordinary point of inflection?
- d For what values of x is the graph concave down?

15 Consider the function $y = 3x^3 + 6x^2 + 8x$ 

- a Find y'' .
- b Find the point of inflection.
- c Is the point of inflection, an ordinary or horizontal point of inflection?
- d For what values of x is the graph concave down?

16 Consider the function $y = x^5 - 3x^2$.

- a Find y'' .
- b Find the exact point of inflection.
- c Is the point of inflection an ordinary or a horizontal point of inflection?
- d For what values of x is the graph concave up?
- e For what values of x is the graph concave down?

17 The first derivative of a certain function is $f'(x) = 3x^2 + 9x$.

- a Determine the interval over which the function is increasing.
- b Determine the interval over which the function is decreasing.
- c Find $f''(x)$.
- d Determine the interval over which the function is concave up.
- e Determine the interval over which the function is concave down.
- f Find the x -coordinate of the maximum turning point.
- g Find the x -coordinate of the potential point of inflection.

Stationary points

18 Consider the function $f(x) = (x - 5)^2 + 3$.

- a Find the x -coordinate of the turning point.
- b Find the value of the second derivative at this point.
- c Hence, classify the stationary point.

19 Consider the function $f(x) = (x - 5)(x + 4)^2$.

- a Find the turning points.
- b Find the second derivative.
- c Determine the nature of the turning points.



20 Consider the function $y = x^4 + 4x^3 + 2$.

- a Find y' .
- b Find y'' .
- c Find the stationary points.
- d Classify the stationary points.
- e Find the points of inflection.
- f Classify each point of inflection as ordinary or horizontal point of inflection.
- g For what values of x is the graph concave up?

21 Consider the function $f(x) = (x - 8)(x - 5)^2$.

- a Find $f'(x)$.
- b Find the turning points.
- c Find $f''(x)$.
- d Classify the turning points.
- e For what values of x is the graph concave down?

22 For each of the following functions:

- | | |
|------------------------------|---------------------------------|
| i Find the turning points. | ii Classify the turning points. |
| a $f(x) = x^3 - 27x - 7$ | b $f(x) = x^3 - 12x - 2$ |
| c $f(x) = 2x^3 + 9x^2 - 24x$ | d $f(x) = x^4 - 2x^2 + 3$ |

23 For each of the following functions:

- | | |
|---|--------------------------------------|
| i Find the x -coordinates of the stationary points. | |
| ii Determine the nature of the stationary points. | |
| a $f(x) = 7x^3 + 8x^2 + 3x + 4$ | b $f(x) = 8(x - 7)^2(x + 2)$ |
| c $f(x) = -\frac{8}{x^2} + 2x$ | d $f(x) = 2x^3 - 30x^2 + 150x - 247$ |
| e $f(x) = 2(x - 5)^3 + 1$ | |

24 Consider the function $f(x) = 5x^3 + 7x^2 + 3x + 6$.

- a Find the x -coordinates of stationary points.
- b Find the x -value of the point of inflection.

